

Narrative Feedback via Earable Interaction to Support Embodied Running Experiences

Yihan Dong
School of Design and Arts
Beijing Institute of Technology
Beijing, China
3120215895@bit.edu.cn

Xipei Ren[†]
School of Design and Arts
Beijing Institute of Technology
Beijing, China
University of Macau
Macao SAR, China
x.ren@bit.edu.cn

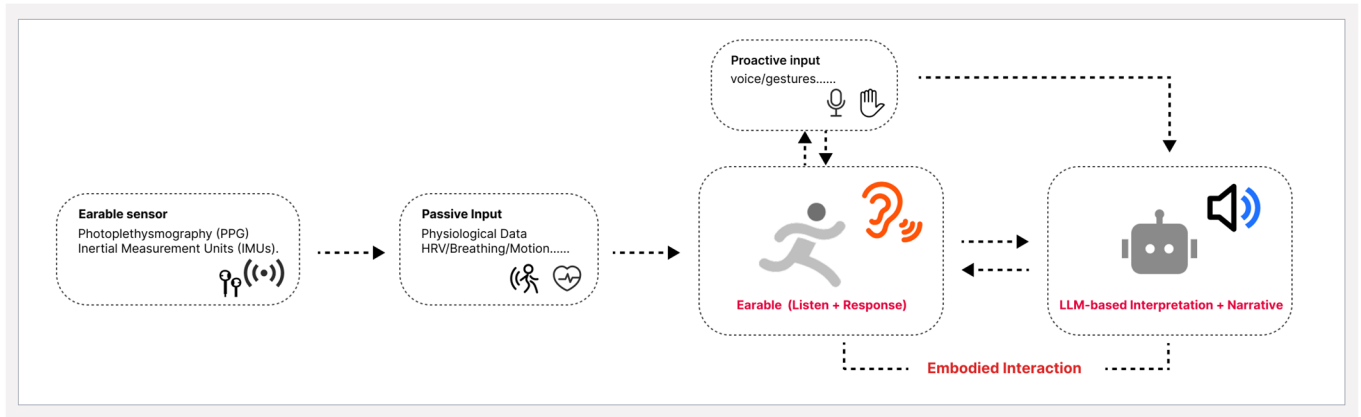


Figure 1: System Interaction Flow

Abstract

This paper proposes a conceptual earable-based interaction system that supports real-time bodily reflection for endurance runners. By integrating multimodal inputs with narrative feedback generated by large language models (LLMs), the system aims to enhance runners' embodied awareness while preserving cognitive and physical flow. We highlight key challenges in sensing accuracy, feedback design, and system adaptability. This work invites the community to explore this approach for enhancing reflection during running and advancing embodied interaction paradigms.

CCS Concepts

• Human-centered computing → Ubiquitous and mobile computing theory, concepts and paradigms.

[†] Corresponding author

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Keywords

Earables; Reflection-in-Action; Embodied Interaction; Large Language Models; Narrative Feedback

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1 Introduction & Motivation

Wearable technologies have become central to endurance sports, enabling runners to track metrics such as pace, cadence, and heart rate. Yet such data-centric feedback often overshadows the runner's own embodied experiences—sensations of strain, emotional states, or subtle bodily cues that guide self-regulation during effort [7]. SportsHCI research has emphasized that overreliance on quantifiable data may inhibit the development of somatic awareness, constraining adaptive decision-making and intrinsic engagement [3] [8].

Despite this, current interactive systems primarily support **reflection-on-action**, offering performance summaries after the

activity ends [2]. These tools rarely engage with the runner during **reflection-in-action**, when shifts in physical or emotional state often occur fluidly within a flow-like experience. Enabling timely self-awareness during effort remains an open challenge.

Sensory earables offer a promising and underexplored opportunity to support reflection-in-action in endurance sports. As everyday devices worn throughout running, earable-based systems now integrate real-time voice input and output, inertial and motion sensing, and in some prototypes, biosignal collection such as heart rate or temperature. Prior research in ubiquitous computing has leveraged earables for on-body gesture interaction [1], facial expression detection [5], and social interaction sensing [4]. However, their use as active mediators for subjective experience, particularly during physical exertion, remains largely unexplored.

This work proposes a conceptual earable-based system that bridges voice input and physiological sensing with low-interruption, narrative feedback generated via large language models (LLMs). Our vision is to support in-situ bodily reflection during running by transforming subjective experience into situated, actionable insight without disrupting the rhythm or focus of the activity.

2 Conceptual System Vision

To facilitate embodied awareness and support reflection-in-action, this conceptual system envisions a closed-loop earable interface that integrates multimodal sensing and narrative language feedback. LLMs are employed not only to translate raw inputs into personalized reflections but also to promote a narrative-based understanding of one's physical state [2]. Rather than issuing explicit performance directives, such feedback seeks to evoke subtle, metaphorical interpretations that better align with athletes' internal sense-making processes [6]. Compared to standard coaching cues that deliver directive commands (e.g., "Pick up the pace", "Relax your shoulders"), narrative feedback encourages attentiveness to one's bodily state rather than compliance with external demands. Prior research highlights that subtle, context-sensitive prompts help athletes cultivate intuitive bodily awareness and confidence in their embodied skills [3]. Within this framing, earables serve as an integrative platform for real-time data acquisition and adaptive feedback generation (See Figure 1).

2.1 Input Modalities

Input is conceptualized along two axes: passive (implicit) and proactive (intentional), forming a classification framework for interaction triggers.

Passive input is derived from continuous physiological and motion sensing using Photoplethysmography (PPG) and Inertial Measurement Units (IMUs). These sensors monitor Heart Rate Variability (HRV), indirectly infer breathing rhythm, and motion features such as head tilt and acceleration. These signals are

continuously analyzed using lightweight algorithms (e.g., decision trees, threshold models) to detect physiological anomalies, such as sustained HRV drops or irregular cadence patterns, which may indicate fatigue. Once such anomalies are identified, feedback is selectively triggered to support bodily awareness and potential adjustment.

Proactive input comes from user-initiated actions, including spontaneous verbal expressions processed by on-device Automatic Speech Recognition (ASR) models such as keyword spotting or lightweight transformer models. Users can also perform lightweight physical gestures, such as tapping the earbud or nodding, detected through IMU-based gesture recognition. These gestures can be self-initiated or employed during interactions to confirm, reject, or request clarification of feedback. When such patterns are identified, the system composes a contextual prompt combining physiological data and recent input, which is forwarded to the LLM for narrative feedback.

2.2 LLM-Driven State Interpretation

An LLM, potentially fine-tuned on running-specific datasets, processes incoming signals to construct interpretable reflections.

For **passive input**, the system generates reflective utterances that metaphorically translate physiological signals. HRV trends are interpreted in context and mapped to naturalistic expressions. For example, it might say, "Your breathing feels like waves breaking, steady and deepening." Alternatively, goal-oriented phrasing can gently guide pacing adjustments without sounding imperative. For instance: "You've been running for 25 minutes. Ease off slightly to let your breath settle."

For **proactive inputs**, the interaction adopts a conversational tone. When users express themselves via speech or gestures, the LLM leverages Natural Language Understanding (NLU) capabilities to provide context-aware responses. For instance, if a user says, "I'm not sure I can keep up," the system might respond with an empathetic suggestion: "You sound tired. Want to slow down a bit together?" Gestural confirmations, such as nods detected by IMUs, can affirm or override the system's response. LLM prompting integrates physiological trends and user input to determine appropriate response tone (e.g., empathetic, metaphorical, instructive) based on user state. A rule-based scaffold may help filter inappropriate or abrupt responses. These styles remain exploratory and dependent on hardware capabilities and user preferences.

2.3 Earable as an Integrated Interface

The earable device plays a central role as both a sensing and real-time feedback node. Compared to screens or wrist-based devices, earables enable more immersive and minimally disruptive engagement during physical activity. The form factor of the device further shapes interaction dynamics. Closed-ear designs can enhance focus and sensorial immersion, while bone-conduction or open-fit designs preserve ambient sound awareness, which is essential for safety during outdoor running.

Moreover, the acoustic qualities of feedback, including tone, pace, and linguistic style, interact with the user's physical and emotional state. These factors influence how reflective content is perceived and internalized. Rather than acting as a passive data collector, the earable is conceptualized as an interpretive interface mediating sensed data and narrative feedback.

3 Research Challenges & Open Questions

Designing input mechanisms that accurately reflect runners' internal states remains a central challenge.

For **passive sensing**, questions remain about how physiological signals—such as heart rate variability, cadence shifts, or respiratory patterns—collected through earable sensors can meaningfully trigger reflective feedback. Determining when and how such signals should initiate LLM-driven responses without misinterpretation is an open research area.

For **active input**, voice remains the most expressive modality, but its use during running raises concerns around timing, cognitive load, and naturalness. If voice is used, how can systems identify meaningful keywords or linguistic cues in unstructured speech and respond supportively without breaking immersion? Could alternative modalities, such as rhythmic patterns, subtle head movements, or tapping gestures, serve as low-effort and naturalistic triggers?

Beyond input modalities, the design of **output feedback** presents its own challenges. A central challenge is aligning the narrative tone with the user's underlying intent, whether motivational, neutral, or reflective, and examining how such tone might adapt over time. In addition to temporal coordination, the linguistic qualities of LLM-generated narrative feedback require careful attention. It is also unclear how users might actively guide or adjust the feedback's tone and style as their preferences evolve. Together, these considerations underscore the need for output design that is both experientially grounded and dynamically co-adaptive.

4 Future Considerations

The integration of LLMs with physiological and voice-based inputs necessitates rigorous attention to data privacy and the risks of generative AI. Biosignals and user-expressed emotional states are inherently sensitive, raising both ethical and regulatory considerations. Moreover, LLMs may generate hallucinated or contextually inappropriate narratives that distort physical feedback or emotional tone, potentially undermining user trust and self-awareness during exertion. To mitigate these issues, future implementations should explore privacy-preserving techniques and empower users to adjust or validate feedback, enhancing its accuracy and relevance.

While this work outlines a conceptual framework, subsequent efforts will focus on building a low-fidelity prototype and conducting in-situ user studies. These exploratory studies will

assess usability, the effectiveness of minimally disruptive narrative feedback, and the system's adaptability to diverse runners' needs in real-world settings. Ethical approval will be obtained from institutional review boards prior to any human studies, ensuring alignment with research standards and robust protection of participant data.

We see this work as a step toward reimagining earables not merely as feedback devices, but as interactive companions that foster bodily awareness and support self-regulation. Ultimately, we invite the community to critically engage with this emerging earable-LLM paradigm by examining how interactive sensing, adaptive language, and support for embodied self-regulation might reshape real-time reflection during endurance activities.

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